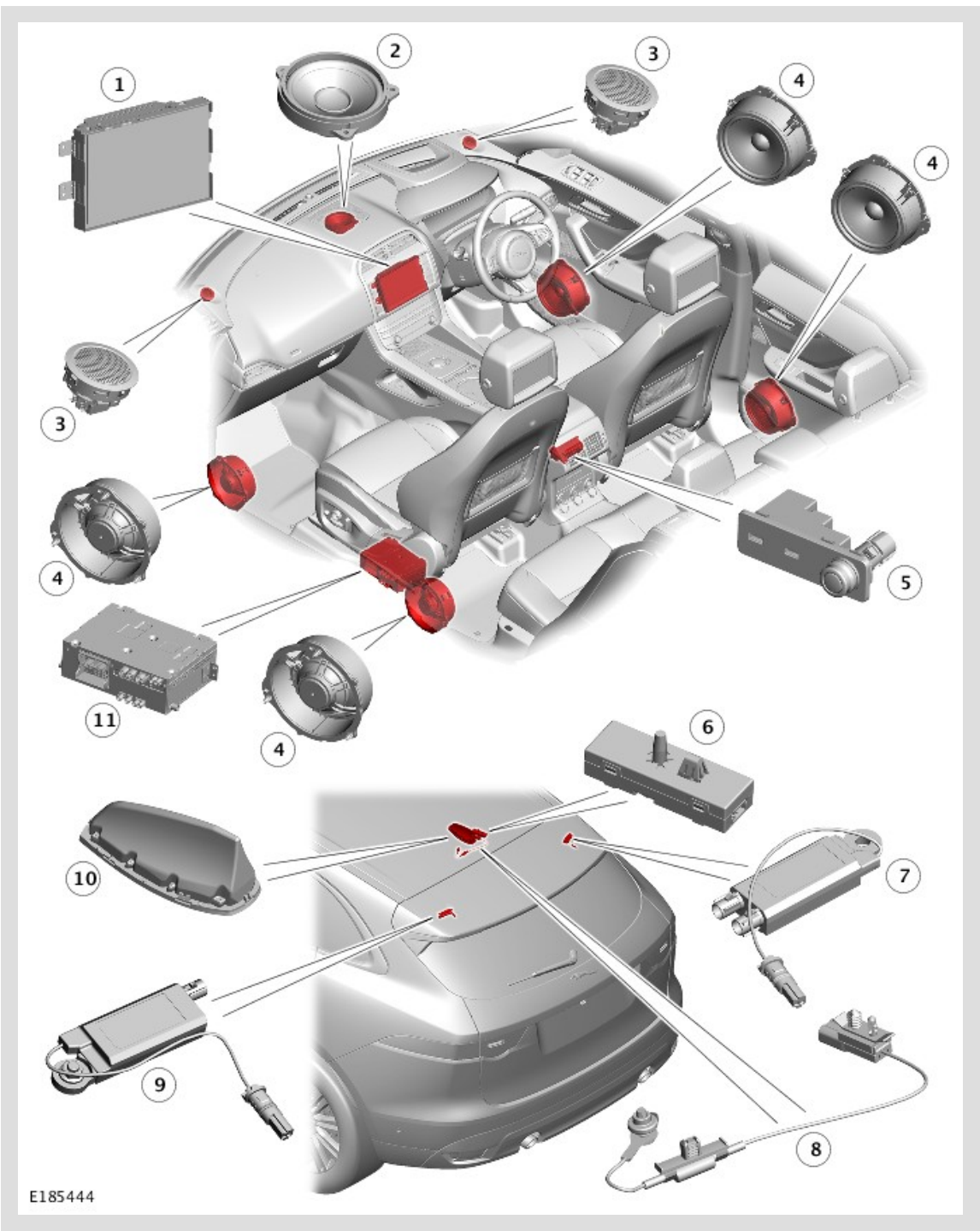


INFORMATION AND ENTERTAINMENT SYSTEM - VEHICLES WITH:
INCONTROL TOUCH PRO

DESCRIPTION AND OPERATION

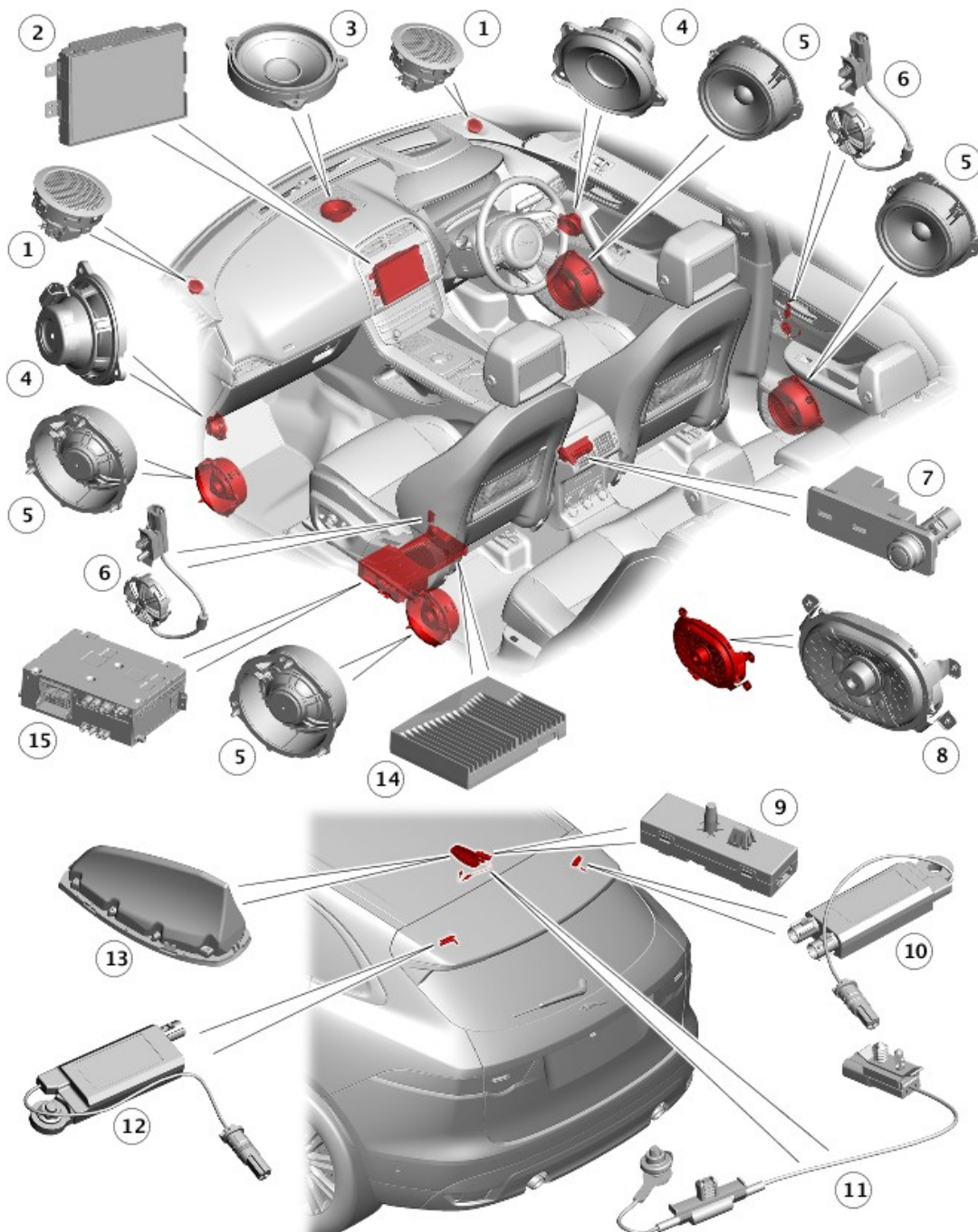
COMPONENT LOCATION - INCONTROL TOUCH



ITEM	DESCRIPTION
1	Touch Screen (TS)

2	Telematics speaker (Russia markets only)
3	Front tweeter speaker (2 off)
4	Bass speaker (4 off)
5	Portable audio interface panel
6	Frequency Modulation 2 (FM) / DAB-3 antenna amplifier
7	Radio Frequency (RF) filter - High mounted stop lamp
8	RF filter - Heated Rear Window (HRW)
9	AM / FM antenna amplifier
10	Roof pod - Global Navigation Satellite System (GNSS) / Digital Audio Broadcast (DAB) / Global System for Mobile communications (GSM) / Satellite Digital Audio Radio Service (SDARS) / external WiFi antenna
11	Audio Head Unit (AHU)

COMPONENT LOCATION - INCONTROL TOUCH MERIDIAN® SOUND

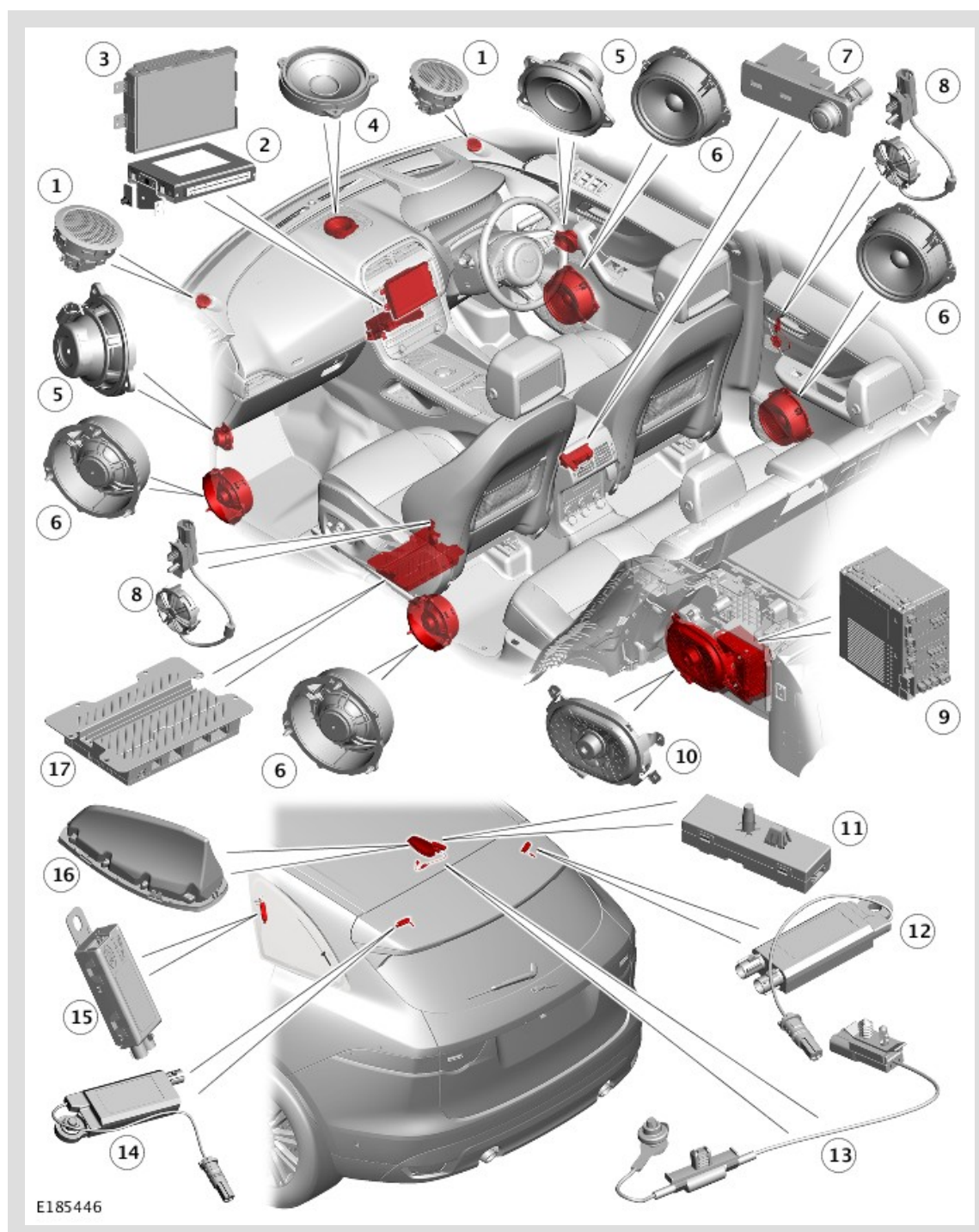


E185445

ITEM	DESCRIPTION
1	Front tweeter speaker (2 off)
2	Touch Screen (TS)
3	Telematics speaker (All markets except Russia)
4	Front mid-range speaker (2 off)
5	Bass speaker (4 off)
6	Rear tweeter speaker (2 off)
7	Portable audio interface panel
8	Sub-woofer speaker
9	Radio Frequency (RF) filter - High mounted stop lamp
10	Frequency Modulation 2 (FM) / DAB-3 antenna amplifier

11	RF filter - Heated Rear Window (HRW)
12	AM / FM antenna amplifier
13	Roof pod - Global Navigation Satellite System (GNSS) / Digital Audio Broadcast (DAB) / Global System for Mobile communications (GSM) / Satellite Digital Audio Radio Service (SDARS) / external WiFi antenna
14	Audio Amplifier Module (AAM)
15	Audio Head Unit (AHU)

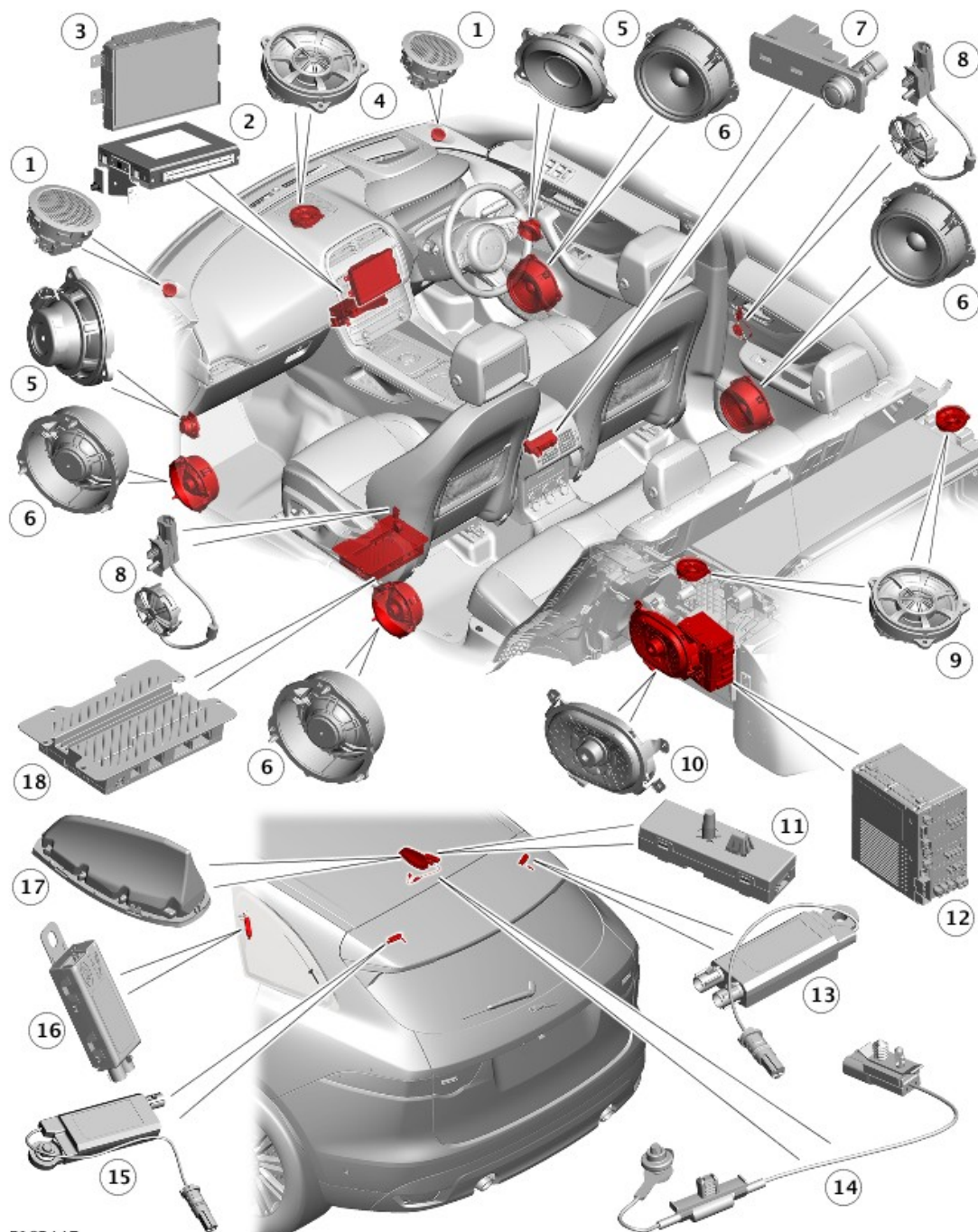
COMPONENT LOCATION - INCONTROL TOUCH PRO MERIDIAN® DIGITAL SOUND



ITEM	DESCRIPTION
1	Front tweeter speaker (2 off)
2	Digital disc player

3	Touch Screen (TS)
4	Telematics speaker (Russia markets only)
5	Front mid-range speaker (2 off)
6	Bass speaker (4 off)
7	Portable media interface panel
8	Rear tweeter speaker (2 off)
9	Infotainment Master Controller (IMC)
10	Sub-woofer speaker
11	Radio Frequency (RF) filter - High mounted stop lamp
12	Frequency Modulation 2 (FM) / DAB-3 antenna amplifier
13	RF filter - Heated Rear Window (HRW)
14	AM / FM antenna amplifier
15	TV and/or DAB 3 antenna amplifier
16	Roof pod - Global Navigation Satellite System (GNSS) / Digital Audio Broadcast (DAB) / Global System for Mobile communications (GSM) / Satellite Digital Audio Radio Service (SDARS) / external WiFi antenna
17	Audio Amplifier Module (AAM)

COMPONENT LOCATION - INCONTROL TOUCH PRO - MERIDIAN® DIGITAL SURROUND SOUND



E185447

ITEM	DESCRIPTION
1	Front tweeter speaker (2 off)
2	Digital disc player
3	Touch Screen (TS)
4	Instrument panel co-axial speaker
5	Front mid-range speaker (2 off)
6	Bass speaker (4 off)
7	Portable media interface panel
8	Rear tweeter speaker (2 off)
9	Luggage compartment co-axial speaker (2 off)
10	Sub-woofer speaker

11	Radio Frequency (RF) filter - High mounted stop lamp
12	Infotainment Master Controller (IMC)
13	Frequency Modulation 2 (FM) / DAB-3 antenna amplifier
14	RF filter - Heated Rear Window (HRW)
15	AM / FM antenna amplifier
16	TV and/or DAB 3 antenna amplifier
17	Roof pod - Global Navigation Satellite System (GNSS) / Digital Audio Broadcast (DAB) / Global System for Mobile communications (GSM) / Satellite Digital Audio Radio Service (SDARS) / external WiFi antenna
18	Audio Amplifier Module (AAM)

OVERVIEW

The audio system is available in four versions. Two of the InControl Touch systems use an Audio Head Unit (AHU) and two of the InControl Touch Pro systems use an Infotainment Master Controller (IMC).

InControl Touch systems are controlled using an 8" Touch Screen (TS) and steering wheel mounted controls.

InControl Touch Pro systems are controlled using a 10.2 inch, 21:9 wide TS and steering wheel mounted controls.

AUDIO HEAD UNIT (AHU) - INCONTROL TOUCH AUDIO SYSTEMS - ALL MARKETS EXCEPT JAPAN AND KOREA

NOTE:

InControl Touch audio systems are not available in Japan and Korea markets.

There are two levels of InControl Touch audio system, both systems use an Audio Head Unit (AHU):

- InControl Touch 8 speaker 80W audio system with integrated 4 channel audio amplifier
- InControl Touch Meridian® Sound 11 speaker 380W audio system with external 12 channel Audio Amplifier Module (AAM) and sub-woofer.

Two versions of the AHU are available:

- Level B AHU features:
 - Integral 4 channel amplifier
 - Amplitude Modulation (AM)/Frequency Modulation (FM)
 - Digital Radio
 - External media player inputs
 - Bluetooth® connectivity
- Level C AHU features:
 - External 12 channel amplifier

- AM/FM
- Digital Radio
- Audyssey MultEQ XT audio tuning system
- External media player inputs
- 'InControl' telematics system
- WiFi hotspot
- Voice recognition
- SMS text display
- Navigation system with Secure Digital (SD) memory card updates
- Bluetooth® connectivity

Wired connectivity is made to the vehicle using a portable audio interface panel or portable media interface panel, depending on system fitted. The level of connectivity is dependent upon the vehicle infotainment equipment level.

InControl Touch systems have a portable audio interface panel which can contain sockets for Secure Digital (SD) card for navigation and voice control, a 3.5mm auxiliary media input socket, a Universal Serial Bus (USB) 1.0/2.0 socket and a 12V accessory socket.

Primary user control of the InControl Touch audio system is via the Touch Screen (TS) which is located in the center of the instrument panel. Control signals from the TS are relayed on an Automotive Pixel Link 2 (APIX2) connection to the AHU . The AHU relays the control signals to the rest of the audio system on a High Speed (HS) Controller Area Network (CAN) infotainment systems bus.

INFOTAINMENT MASTER CONTROLLER (IMC) - INCONTROL TOUCH PRO AUDIO SYSTEMS - ALL MARKETS

There are two levels of InControl Touch Pro audio system, both systems use the Infotainment Master Controller (IMC):

- InControl Touch Pro Meridian® Digital Sound 11 speaker 380W Meridian® audio system with external 12 channel Audio Amplifier Module (AAM) and sub-woofer speaker
- InControl Touch Pro Meridian® Digital Surround sound 17 speaker 825W Meridian® Surround audio system with external 16 channel AAM and sub-woofer speaker.

The InControl Touch Pro audio systems include the following features:

- 8 single view and 10 inch single and dual view Touch Screens (TS) available
- HD Radio™ (NAS only)
- Satellite Digital Audio Radio Service (SDARS) (NAS only)
- Digital Audio Broadcast (DAB)
- 60GB media storage Solid State Drive (SSD)
- Multiple Bluetooth® connectivity
- Personalised home screens

- SSD based navigation with connected services and features
- Multiple port/device support 2 x USB 3.0, HDMI, MHL
- Digital TV - can view two concurrent channels
- Amplitude Modulation (AM)/Frequency modulation (FM) tuners
- Auto switching from DAB to FM (ROW), HD Radio™ to FM (NAS only) when signal degrades
- Multiple antenna for DAB/HD Radio™ and FM
- Media library - enhanced support for USB flash drive
- Incontrol Telematics - WiFi and Apps
- 8 inch single view or 10 inch single and dual view Touch Screen (TS) with multi touch gestures (Swipe - Flicking - Panning - Pinching - Zoom)
- Integrated voice interaction (17 languages, 3 of which support natural input: UK/US English, Chinese (Mandarin))
- Voice Control
- Digital disc player for ripping of music Compact Discs (CDs) or playing of Digital Versatile Discs (DVDs).

Wired connectivity is made to the vehicle using a portable media interface panel. The level of connectivity is dependent upon the vehicle infotainment equipment level.

The InControl Touch Pro systems portable media interface panel contains two USB 3.0 sockets, one High-Definition Multimedia Interface (HDMI) / Mobile High-Definition Link (MHL) socket, one Subscriber Identification Module (SIM) card socket and a 12V accessory socket.

Primary user control of the InControl touch Pro audio system is via the Touch Screen (TS) which is located in the center of the instrument panel. Control signals from the TS are relayed on an Automotive Pixel Link 2 (APIX2) connection to the IMC . The IMC relays the control signals to the rest of the audio system on a High Speed (HS) Controller Area Network (CAN) comfort systems bus.

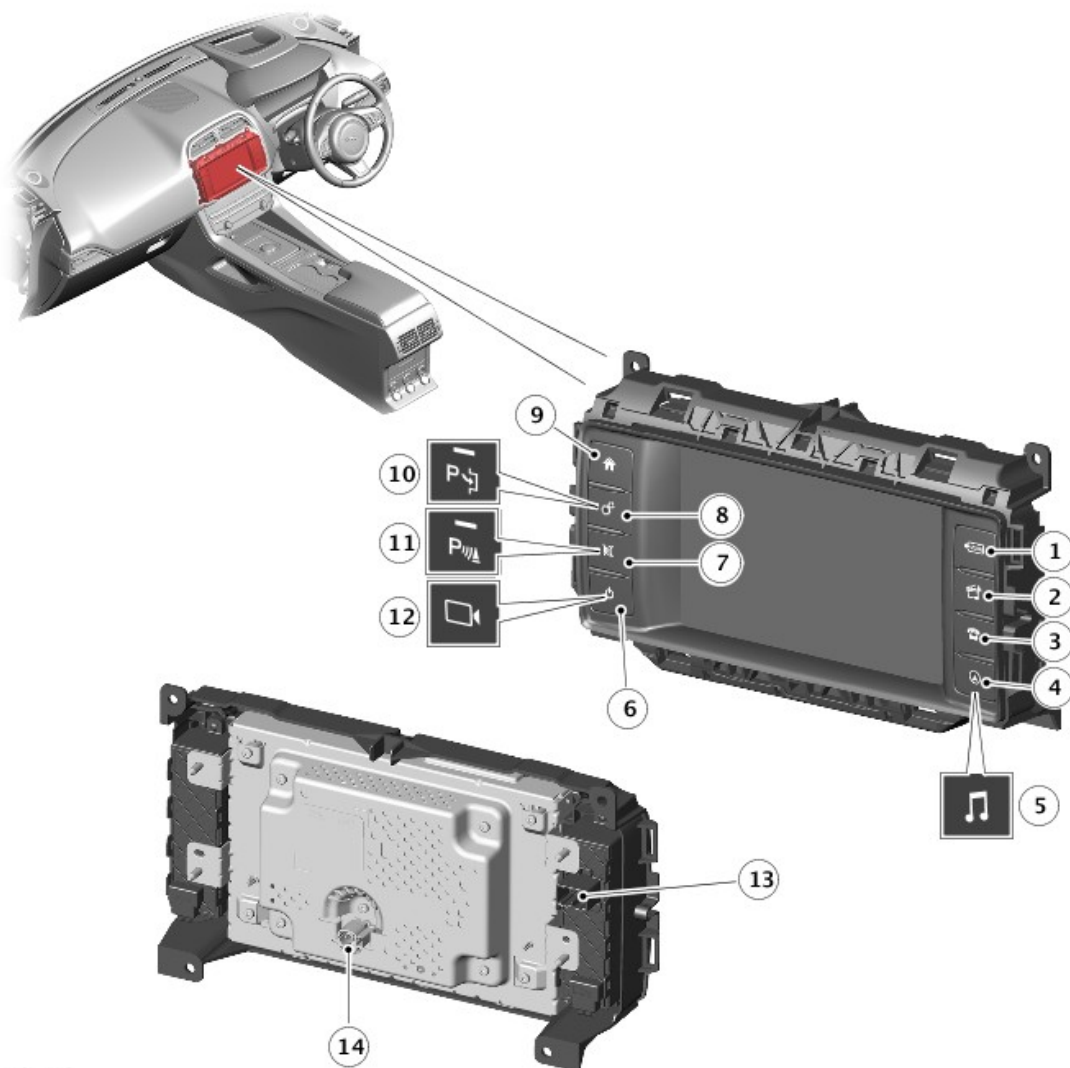
DESCRIPTION

TOUCH SCREEN (TS)

Touch Screen - Audio Head Unit - InControl Touch

NOTE:

Switch functions vary depending on vehicle specification



E185448

ITEM	DESCRIPTION
1	'Mode' switch
2	'Media System' switch
3	'Telephone System' switch
4	'Audio Settings' switch
5	'Navigation' switch
6	Infotainment system On/Off switch
7	Audio mute switch
8	'General Settings' switch
9	'Home Menu' switch
10	'Park Assist' switch
11	'Front Parking Aid' On/Off switch (if fitted)

12	'Camera' switch
13	Connector - Touch screen switchpack
14	Connector - Touch screen

The 8 inch 800 X 480 resolution, single view Touch Screen (TS) is located in the center of the instrument panel. The TS comprises an 8 inch color, touch sensitive display, with a TS switchpack located on either side.

The TS switchpack differs depending on system specification; for example, if parking aid or navigation are specified on the vehicle, these switches will replace 'Audio Settings' or 'Mute' functions respectively. The functions can still be accessed via the home menu using the TS.

The TS is the primary user interface of the audio system. It communicates with the other components of the audio/infotainment system on the medium speed Controller Area Network (CAN) comfort systems and allows control of the audio system and other system functions from a single point. No configuration procedure is required if the TS or TS switchpacks are replaced.

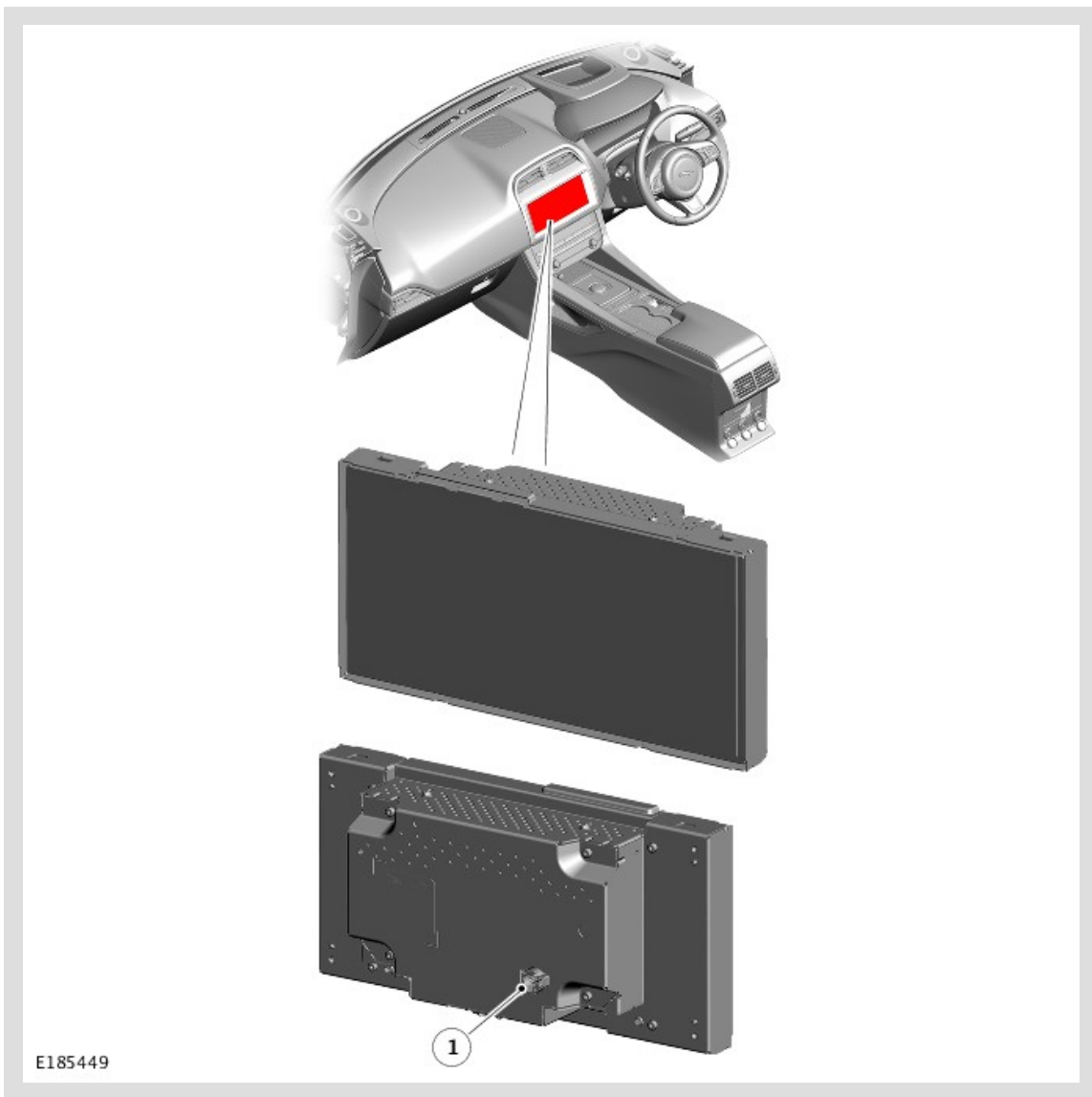
Calibration of the TS using approved Jaguar diagnostic equipment enables updates to be downloaded as new technology becomes available or any fault concerns require software updates.

The rear of the TS has a single 6 pin connector for power, ground and APIX2 connections.

The TS switchpack comprises 8 switches. The function of the switches can differ dependant on the infotainment system installed in the vehicle.

The TS switchpack has a single 12 pin connector which provides electrical connections to and from the switches. The connector provides switch power supplies, switch illumination supplies and output from the switches via a resistive ladder. The outputs from the Parking Aid and Park Assist switches are passed directly to the Parking Aid Control Module (PACM).

Touch Screen - InControl Touch Pro Audio Systems - All markets



ITEM	DESCRIPTION
1	APIX2 connector

Two variants of the Touch Screen (TS) are available for the InControl Touch Pro systems:

- 10.2 inch, 1280 x 542 resolution single view
- 10.2 inch, 2560 x 542 resolution dual view (Not available for NAS markets)

The InControl Touch Pro TS do not have the switch packs on either side of the screen. The switches are replaced with soft key touch icons located in the screen image.

The TS is located in the center of the instrument panel and is connected to the Infotainment Master controller (IMC) by a single APIX2 connection. The APIX2 connection provides power and ground connections and also data input and output. Power and ground is provided by the IMC.

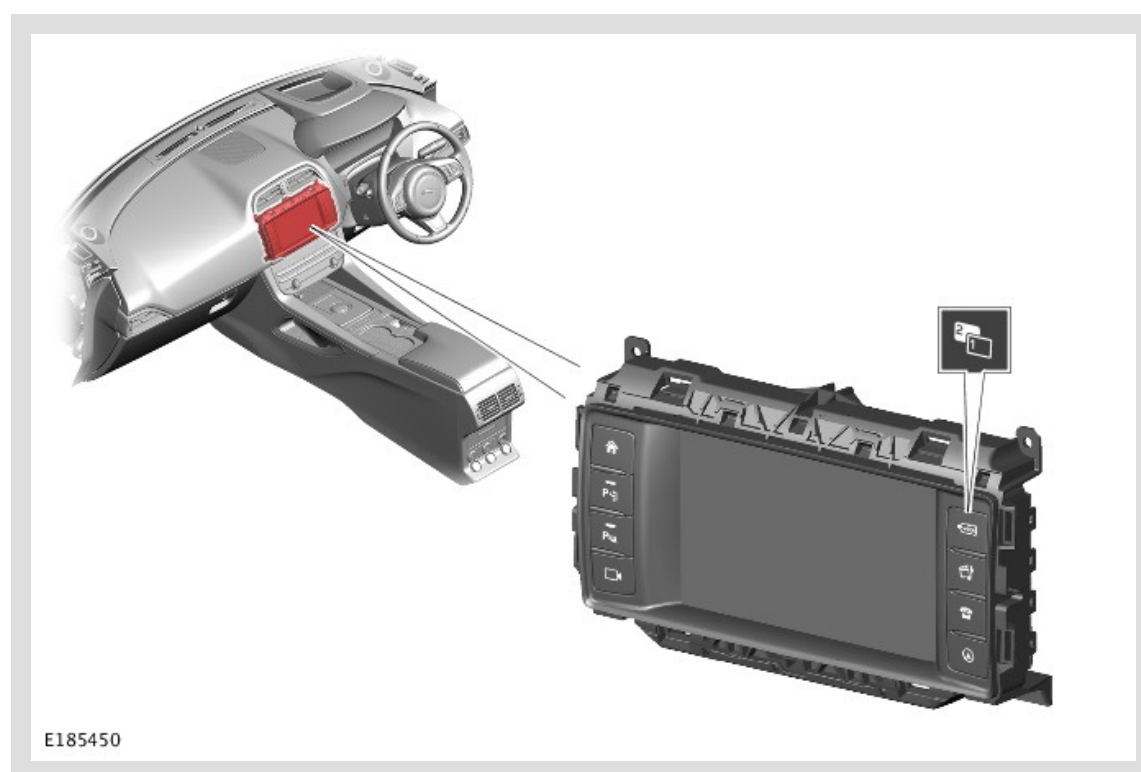
Two versions of the TS are available; single view and dual view.

The TS is the primary user interface of the audio system. It communicates with the other components of the audio/infotainment system on the APIX2 connection via the IMC and allows control of the audio system and other infotainment systems from a single point.

Calibration of the TS using approved diagnostic equipment enables updates to be downloaded as new technology becomes available or any fault concerns require software updates.

Dual View Touch Screen (TS) - where fitted

Dual View Touch Screen (TS) - InControl Touch



Dual View Touch Screen (TS) - InControl Touch Pro



The dual-view TS enables the passenger and driver to view completely different images from their respective seating positions. This technology has provided a solution for the legal issues attached to viewing moving images whilst the vehicle is in motion. It is not possible for the driver to view moving images with an active speed signal, but the passenger can view DVD and TV independent of the driver.

Vehicles fitted with a dual view TS are supplied with a headphone transmitter located in the overhead console and one pair of infra-red wireless headphones. This allows the front seat passenger to view and listen to music, DVD and TV without disturbing the driver.

NOTE:

Due to legislation the NAS markets do not receive this option. A single view display is available in these markets.

The dual-view TS uses Parallax Barrier Shutter Technology to alternately hide and reveal columns of pixels to the left and right hand views of the screen. The display comes with a specially designed anti-glare coating to help prevent sunlight bleaching.

To access a TV or video image when the vehicle is in motion and single view is selected, the dual view switch on the TS switchpack or the dual view soft key should be pressed by either the driver or the passenger. This will then change the TS to dual-view mode allowing the passenger to view TV or video, but not the driver. A second press of the switch or soft key will change the TS back to single view.

Once dual-view has been selected, the driver can change the current screen without affecting the passengers view by the source on the TS.

The audio system can only broadcast one audio source. Therefore, the TV / video source that is current for the passenger will also be the audio the driver can hear. The passengers can choose to use headphones to listen to the sound source accompanying the TV / video. This allows the driver to listen to a different audio source or navigation commands via the vehicle speaker system.

The driver's view is also event driven, for example, if reverse gear were to be selected, the rear view camera will be displayed automatically, overriding the currently displayed information.

Clock

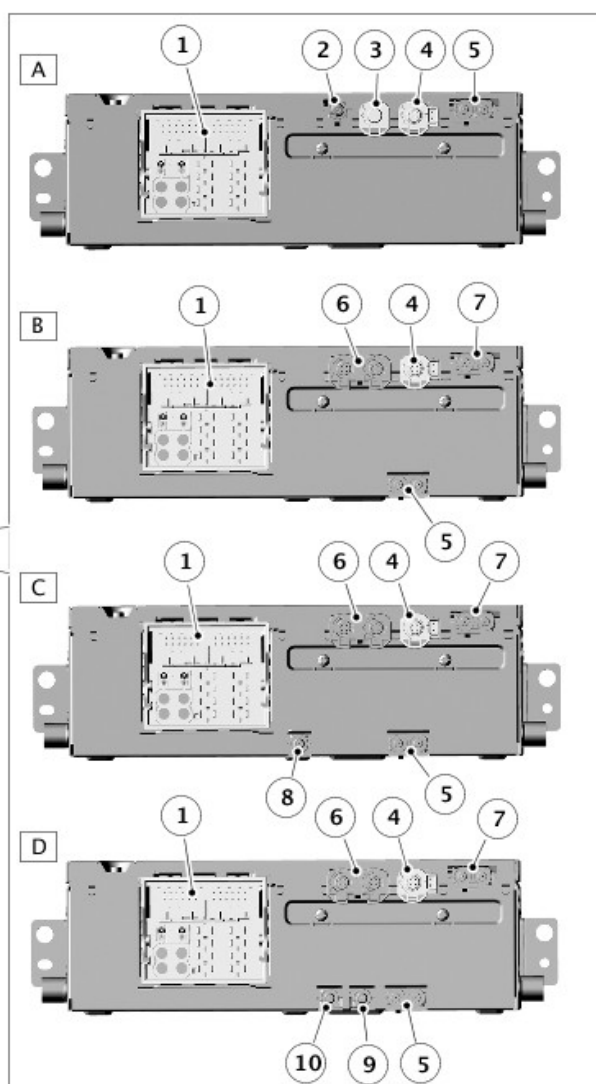
The Body Control Module/Gateway Module (BCM/GWM) assembly contains the master clock functionality. Other vehicle infotainment system modules that require clock functionality use the time supplied from the BCM/GWM assembly.

The clock is available to any control module that is connected to an interconnecting bus system.

The clock display is configurable to show in AM / PM or 24 hour format. Midnight is shown as 12:00AM or 00:00 respectively. The default condition, if not specified, after power on or delivery, should default to 12:00PM or 00:00. Depending upon the market set, the clock will default to either 12 or 24 hour format.

The time is adjusted from the TS. Under conditions when any bus system could be asleep or shut down, the TS does not allow clock adjustments.

AUDIO HEAD UNIT (AHU) - INCONTROL TOUCH AUDIO SYSTEMS



E167297

ITEM	DESCRIPTION
A	Level 'B' AHU - All markets
B	Level 'C' AHU with navigation - All markets
C	Level 'C' - Not used
D	Level 'C' AHU with navigation and DAB - European markets only
1	Connector - 40 pin

2	Connector - Camera
3	Connector - USB
4	Connector - Touch Screen APIX2
5	Connector - AM/FM antenna input
6	Connector - USB
7	Connector - GPS and camera
8	Connector - Not used
9	Connector- DAB-III (European markets only)
10	Connector - DAB-L (European markets only)

The Audio Head Unit (AHU) is located in the center of the instrument panel, below the TS. The AHU is the main infotainment controller for the InControl Touch audio system.

Four levels of AHU are available depending on market and vehicle specification:

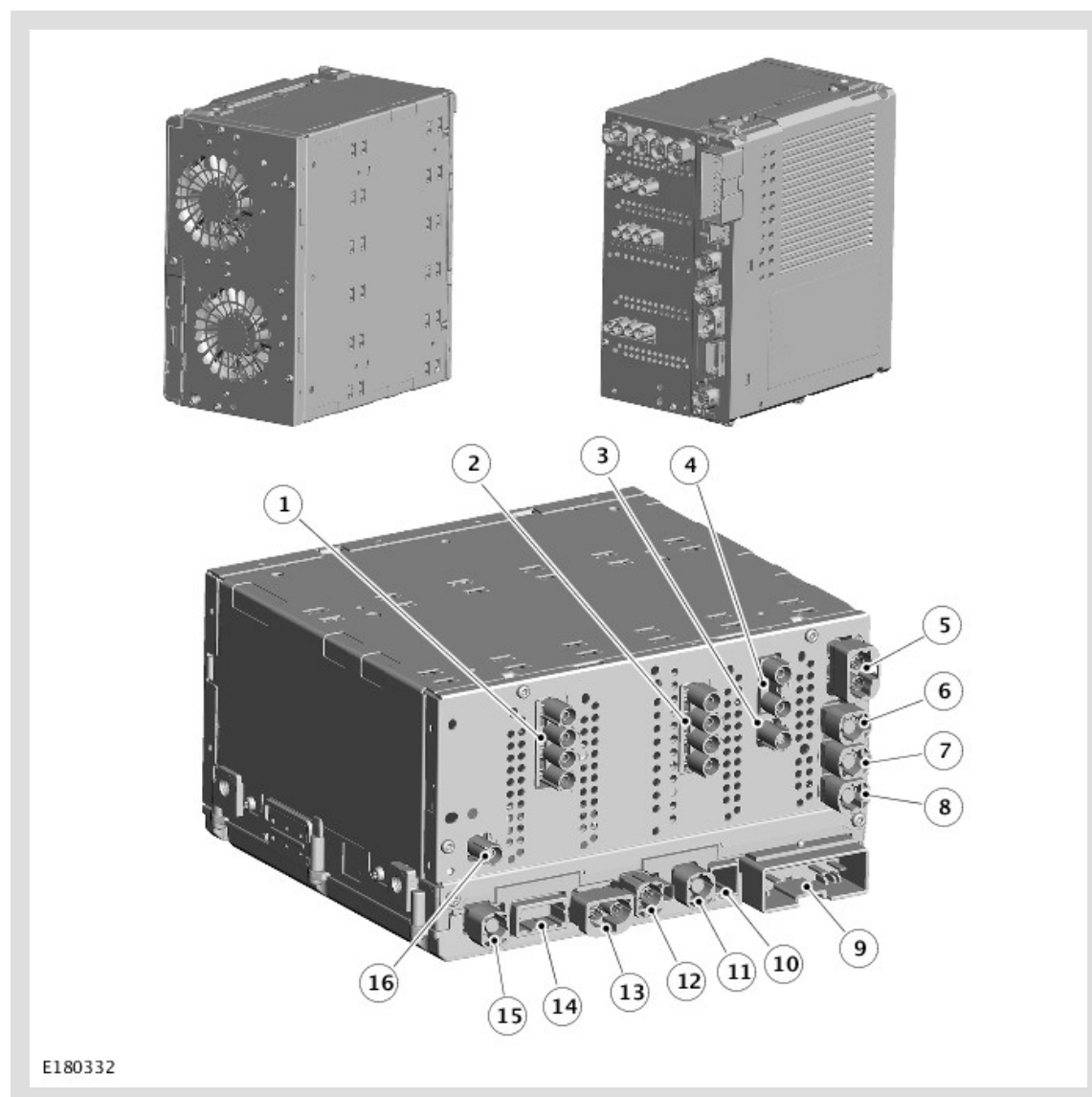
FEATURE	AUDIO HEAD UNIT LEVEL B - ALL MARKETS	AUDIO HEAD UNIT LEVEL C - ALL OTHER MARKETS	AUDIO HEAD UNIT LEVEL C - EUROPE	AUDIO HEAD UNIT LEVEL C - NAS
AM/FM radio Diversity	Yes	Yes	Yes	Yes
DAB radio	No	Yes	Yes	Yes
4-channel internal amplifier	Yes	No	No	No
12-channel external amplifier	No	Yes	Yes	Yes
Bluetooth® phone system	Yes	Yes	Yes	Yes
Bluetooth® audio streaming	Yes	Yes	Yes	Yes
Live phonebook access	Yes	Yes	Yes	Yes
Voice recognition	No	Yes	Yes	Yes
SMS Text display	No	Yes	Yes	Yes
SD Card Global Navigation Satellite System	No	Yes	Yes	Yes
Telematics - E-Call/B-Call Mute only	Yes	Yes	Yes	Yes
Telematics - WiFi/Internet	No	Yes	Yes	Yes
3.5mm Auxiliary input	Yes	Yes	Yes	Yes
USB Input - Media player/Audio /Video/iPod Audio/USB Charging	Yes - 0.5A charging	Yes - 2.1A charging	Yes - 2.1A charging	Yes - 2.1A charging

The AHU receives a 12V fused power supply from the Battery Junction Box (BJB) and the Quiescent Current Control Module (QCCM).

Medium speed Controller Area Network (CAN) Comfort system bus connections provide communication with other system control modules, for example; inputs from the Parking Aid Control Module (PACM) for sound generation and display on the TS.

The AHU uses APIX2 which is a new protocol for video output to the Touch Screen (TS). APIX2 is a high speed digital serial link capable of communicating up to 3 Gbit/second of digital video data. The serial link can support simultaneous video, audio and data transmission from the AHU and the TS.

INFOTAINMENT MASTER CONTROLLER (IMC) - INCONTROL TOUCH PRO AUDIO SYSTEM



ITEM	DESCRIPTION
1	Television (TV) antenna connector sockets (Not NAS)
2	AM/FM/DAB/DAB+/VICS antenna connector sockets
3	GPS antenna connector socket
4	Bluetooth® and Wi-Fi antenna connectors
5	Infotainment Slave Controller (ISC) connector socket (intro 17MY)
6	Audio Amplifier Module (AAM) connector socket
7	Telematics Control Module (TCU) connector socket
8	Diagnostics socket (Not used for service)
9	Primary battery power, ground, Controller Area Network (CAN) bus, microphone input, IMC cooling fan control,

	Local Interconnect Network (LIN) Steering Wheel Module (SWM) connector socket
10	Camera Control Module (CCM) connector socket
11	Instrument Cluster (IC) connector socket
12	Touch Screen (TS) Automotive Pixel Link 2 (APIX2) connector socket
13	USB 3.0 Repeater connector socket - connected to the portable media interface panel
14	Portable media interface panel connector socket - HDMI/MHL and SIM
15	USB 2.0 Hub connector socket - connected to digital disc player and/or the electronic toll collection module (Japan only)
16	Satellite Digital Audio Radio Service (SDARS) connector socket (NAS only)

The Infotainment Master Controller (IMC) is located on the left side of the luggage compartment.

Three levels of IMC are available depending on market and vehicle specification:

- ROW without Television (TV)
- ROW with TV
- NAS Satellite Digital Audio Radio Service (SDARS) without TV.

The IMC is equipped with a high speed Intel® Quad Core 1.9 GHz Atom™ processor with a 4 Gigabyte (GB) Random Access Memory (RAM) combined with a fast access 60GB Solid State Drive (SSD).

The IMC has the following integrated features:

- AM/FM diversity
- DAB radio (ROW)
- SDARS/HD radio™ (NAS only)
- Digital TV (Not NAS or Hong Kong)
- Bluetooth® mobile phone and audio streaming
- InControl Apps
- Voice recognition
- Wi-Fi receiver
- Solid State Drive (SSD) navigation
- 2x USB 3.0 input ports
- 1x HDMI/MHL input port
- 1x SIM card input port
- 60GB SSD storage (10GB user media storage).

When the vehicle systems become active, the IMC is woken up by CAN bus activity. The IMC then turns on the TS.

The IMC incorporates an Amplitude Modulation (AM)/Frequency Modulation (FM) tuner. The following market dependent systems are available:

- ROW (AM/FM/DAB band L/3)
- NAS (AM/FM/HD Radio™/SDARS)

A new feature of radio reception is the ability of the system to automatically switch from DAB to FM (ROW markets) and FM to HD Radio™ (NAS markets), providing a consistent signal when driving between regions.

The IMC radio tuner provides storage of 6 preset stations for the FM and AM frequencies.

The radio tuner also incorporates the following radio functions:

- AM Auto Store (AST)
- FM station list
- Presets - 6 per frequency
- Traffic announcements (TA) – Europe only
- Radio Data System (RDS) functions:
 - Station name
 - Radio text
 - PTY
 - Traffic Announcements (TA) (Not NAS)
 - AF switching (Not NAS)
 - REG to lock to regional/local broadcast (Not NAS)
 - Seek station
 - Tune up/down.

All radio antennas are routed into the IMC via antenna amplifiers. The IMC supplies a power output to the AM/FM antenna amplifier.

The IMC is subject to the power management of the Quiescent Current Control Module (QCCM).

Calibration of the IMC using approved diagnostic equipment enables updates to be downloaded as new technology becomes available or any fault concerns require software updates. If the IMC is replaced it must be configured as a new module using approved diagnostic equipment.

IMC Cooling

The IMC is connected to a duct which draws air through a vent in the left luggage compartment side trim.

Temperature control within the IMC is necessary to ensure optimum performance. IMC temperatures are monitored constantly and controlled by software within the IMC. The temperature assessment software is integrated into the IMC.

At low ambient temperatures of between -30°C and -20°C (-22°F and -4°F) a low temperature condition will be initiated with restricted functionality and access to the SSD is prevented. The following functions will be disabled:

- Media library
- Voice control
- Navigation
- Language settings.

During the low temperature condition a message 'System is warming up' will be displayed in the TS if the navigation soft key is operated. When the ambient temperature increases to more than -20°C (-4°F) the IMC will initiate the SSD without a full re-boot of the system.

At high ambient temperatures of more than 70°C (158°F) a message will be displayed 'System shutting down in 2 minutes' in the TS. If this message appears it could indicate a fault with the IMC or blockage of the ducting.

DIGITAL DISC PLAYER



The digital disc player has the capability to play a number of file formats for both audio and video. The digital disc player can play back standard definition CD/DVD video discs only.

The player is connected to the USB 2.0 Hub. Data is transmitted from the USB 2.0 hub directly to the IMC.

The IMC logs Diagnostic Trouble Codes (DTCs) for faults generated by the digital disc player. When replaced the digital disc player is calibrated, using approved diagnostic equipment.

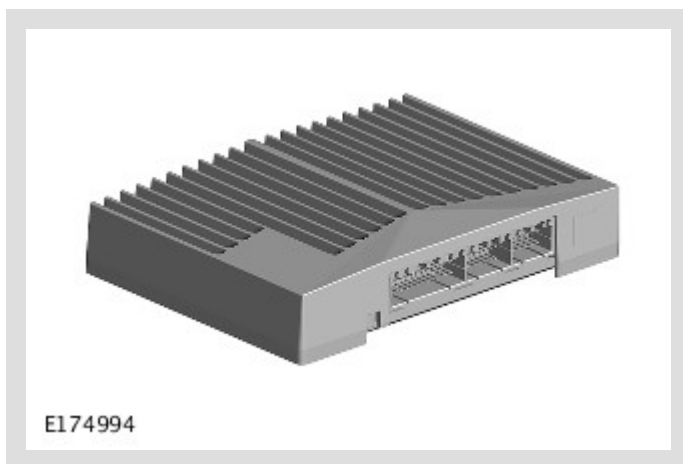
AUDIO AMPLIFIER MODULE (AAM)

Systems with an external Audio Amplifier Module (AAM) have the AAM located below the front left seat. The AAM is secured with a bracket to the vehicle floorpan.

The audio systems have three AAM options, dependent on level of audio system fitted:

- InControl Touch 11 speaker 380W Meridian® audio system with external 12 channel AAM
- InControl Touch Pro 11 speaker 380W Meridian® Digital Sound audio system with external 12 channel AAM
- InControl Touch Pro 17 speaker 825W Meridian® Surround audio system with external 16 channel AAM.

InControl Touch Meridian® 11 Speaker System



The InControl Touch Meridian® Sound 11 speaker system uses an AAM specific to the system. The AAM has a power output of 380W.

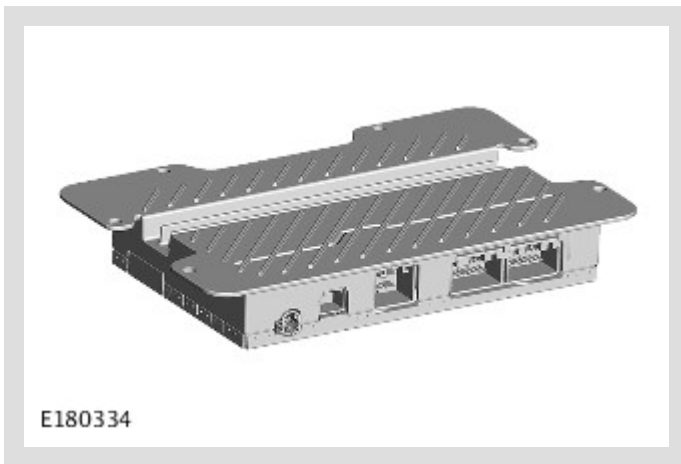
The AAM has three connectors which provide for the following:

- power supply and ground
- Hardwired connections to the vehicle speakers
- High speed CAN infotainment systems bus between the AHU and the AAM

The AAM processes signals from other system modules for output via the vehicle speakers.

Power supply is received from the Rear Junction Box (RJB).

InControl Touch Pro Meridian® Digital Sound 11 Speaker System



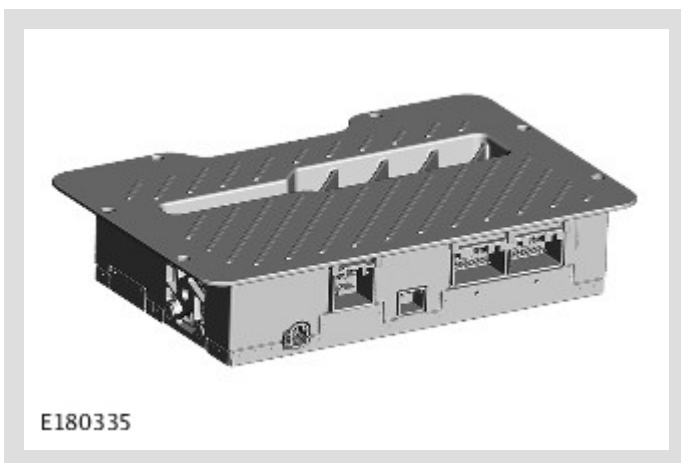
The InControl Touch Pro 11 speaker Meridian® Digital Sound system uses an AAM specific to the system. The AAM has a power output of 380W.

The AAM has five connectors which provide for the following:

- Power supply and ground
- Hardwired connections to vehicle speakers
- Output to headphone transmitter
- BroadR-Reach® ethernet from IMC.

Power supply is received from the RJB.

InControl Touch Pro 17 Speaker Meridian® Digital Surround System



The InControl Touch Pro 17 speaker Meridian® Surround Sound system uses an AAM specific to the system. The AAM has a power output of 825W.

The AAM has five connectors which provide for the following:

- Power supply and ground
- Hardwired connections to vehicle speakers
- Output to headphone transmitter

- BroadR-Reach® ethernet from IMC.

Power supply is received from the RJB.

LEFT STEERING WHEEL SWITCHPACK



ITEM	DESCRIPTION
1	Menu/OK switch (instrument cluster menu control)
2	Volume increase switch
3	Telephone answer/end call switch
4	Voice control switch
5	Mode switch
6	Seek, next radio preset, next track, scroll up telephone list switch
7	Volume decrease switch
8	Seek, previous radio preset, previous track, scroll down telephone list switch

The left steering wheel switchpack is located on the left side of the steering wheel.

The switches are a resistive ladder type which apply a different voltage via a Local Interconnect Network (LIN) bus to the AHU or the IMC in response to different switches being pressed.

The left steering wheel switchpack controls the following switch functions:

- MODE - Press repeatedly to scroll through all audio/video sources
- » Short press up:
 - to select the next radio preset
 - to select the next track on chosen audio source
 - when telephone is in use, press to scroll up lists of calls or phonebook entries.
- « Short press down:
 - to select the previous radio preset
 - to select the previous track or start of current track on chosen audio source
 - when telephone is in use, press to scroll down lists of calls or phonebook entries.
- With radio manual seek mode activated, further short presses will change the frequency in single increments. A further long press will scan forwards through the current waveband until the switch is released
- » Long press up:
 - to auto seek up the frequency to the next radio station.
- « Long press down:
 - to auto seek down the frequency to the next radio station.
- Volume increase for any audio source
- Volume decrease for any audio source.

The MENU/OK switch is for use with the instrument cluster menu. For additional information, refer to: Instrument Cluster (413-01 Instrument Cluster, Description and Operation).

For further details on the telephone switch refer to Cellular Phone. For additional information, refer to: Cellular Phone (415-01 Information and Entertainment System, Description and Operation).

For further details on the voice switch refer to Voice Control. For additional information, refer to: Voice Control (415-01 Information and Entertainment System, Description and Operation).

PORTABLE AUDIO INTERFACE PANEL - INCONTROL TOUCH SYSTEMS



ITEM	DESCRIPTION
A	For InControl Touch systems with Navigation, voice control or telephone
B	For InControl Touch system with Level B AHU

The portable audio interface panel is located in the front floor console stowage compartment.

Two versions of the portable audio interface panel are available dependent on vehicle specification. The versions have combinations of a SD memory card slot, a 3.5mm auxiliary media input socket, a Universal Serial Bus (USB) 1.0/2.0 socket and a 12V accessory socket.

InControl Touch Audio Systems with SD Memory Card

A version of the panel is for the InControl systems for vehicles with navigation system. On these vehicles the portable audio interface panel includes an SD memory card slot for navigation system, voice control and telephone software. For additional information, refer to: Navigation System (415-01 Information and Entertainment System, Description and Operation) / Voice Control (415-01 Information and Entertainment System, Description and Operation).

NOTE:

The SD memory card must not be removed or the files modified. If the card is removed, the navigation, voice control and telephone will be inoperable. The SD memory card slot cannot be used for media files. Installation of any external content or modification of the files on the SD memory card can corrupt the SD memory card and cause voice control, telephone and navigation systems to be inoperable.

Removal and re-installation of the SD memory card when the ignition is on (power mode 6) will cause voice control, telephone and navigation systems to be inoperable. To re-initialise the SD memory card, switch the ignition to 'off' and lock and arm the vehicle (power mode 6). Install the SD memory card in the slot in the portable audio interface panel and switch the ignition to 'on' (power mode 6). The card will be detected and voice control and navigation will operate correctly.

Universal Serial Bus (USB)

The USB socket is connected directly to a USB socket on the Audio Head Unit. The USB allows data transfer between the socket from an external media device. The socket also provides a 5V charging voltage at a current of 0.5A for level B AHU and 2.1A for level C AHU. The USB power supply originates from a fused supply from the Rear Junction box (RJB) and is controlled by the ignition relay. The USB charging supply is available after ignition off (power mode 0) for a pre-determined period while the ignition relay contacts are open.

General Media Information

Devices that can be connected to the portable audio interface panel include:

- USB mass storage devices (for example a memory stick). Devices must use FAT or FAT32 file format.
- iPod® (iPod Classic, iPod touch, iPhone and iPod Nano are supported - full functionality for older devices cannot be guaranteed). iPod Shuffle functionality cannot be guaranteed.
- Auxiliary device (personal audio, MP3 players).
- Devices with Bluetooth® connectivity. Devices must support A2DP and AVRCP Bluetooth® protocols).

When connecting a portable storage device, select AUX on the TS to select that device input. Depending on the device connected, many of the audio controls on the TS and steering wheel switches can be used.

NOTE:

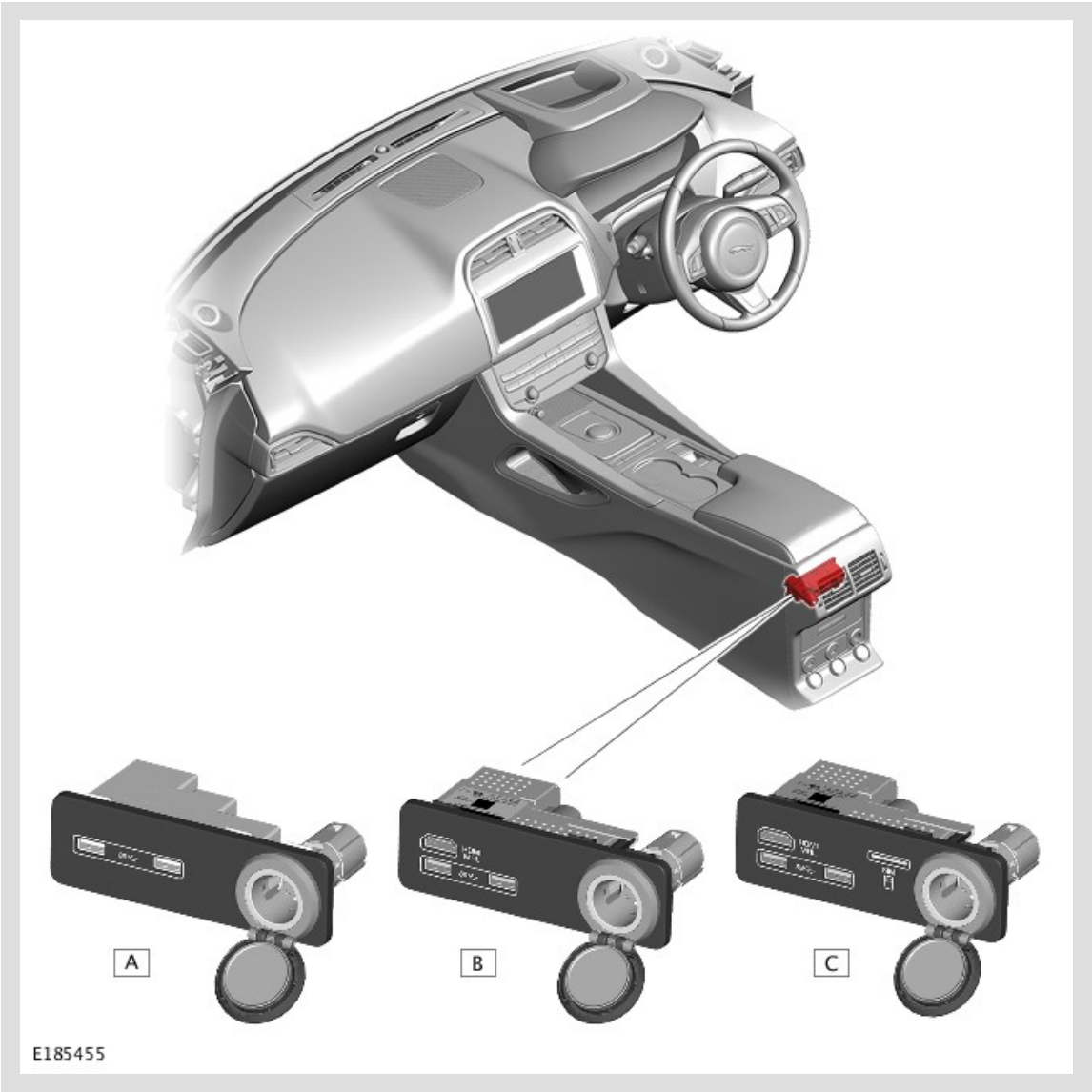
It is recommended to disconnect an iPod when leaving the vehicle. Failure to do so may result in the iPod battery discharging.

To maximise playback quality, it is recommended to use lossless compression for any media files on iPod. Failing this, it is recommended that compressed files utilise a minimum bitrate of 192 kilobits per second (kbps) (a higher bitrate is strongly recommended for increased quality playback).

For media files delivered via USB or MP3 players, the highest compression rate supported is 320 kbps. If anything less than 128 kbps is used, Digital Signal Processing (DSP) functionality may be lost.

Some MP3 players may have a file system that is not supported by the AHU. To use an MP3 player, it must be set to USB Removable Device or Mass Storage Device mode. Only music that has been added to the device in this mode can be played.

PORTABLE MEDIA INTERFACE PANEL - INCONTROL TOUCH PRO SYSTEMS



ITEM	DESCRIPTION
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A	Standard panel - two USB 3.0 sockets
B	InControl Apps panel - two USB 3.0 sockets and one HDMI/MHL socket
C	InControl Apps Connect Pro panel - two USB 3.0 sockets, one HDMI/MHL socket and SIM card socket

The portable media interface panel is located in the front floor console stowage compartment. The portable media interface panel differs from the portable audio interface panel used on the InControl Touch systems in that image data can be passed from a portable media device to the IMC.

Three versions of the portable media interface panel are available dependent on vehicle specification. The versions have combinations of two USB 3.0 sockets, one High Definition Multimedia Interface (HDMI) / Mobile High-Definition Link (MHL) socket, a SIM card socket and a 12V accessory socket.

The standard fitting portable media interface panel includes two USB 3.0 sockets (which also support USB 2.0) and are connected to the IMC through a USB 3.0 signal booster.

The portable media interface panel, fitted with the InControl Apps option, includes two USB 3.0 sockets and a single HDMI/MHL port.

HDMI supports multiple devices for audio/video playback on the Touch Screen (TS). MHL provides charging for supported devices.

Data transfer from the HDMI/MHL port is direct to the IMC, using an APIX2 high speed digital serial link.

The portable media interface panel, fitted with the InControl Connect Pro option, includes two USB 3.0 sockets, a single HDMI/MHL socket and a SIM card socket (micro SIM) for data connectivity.

Data transfer from the SIM card is direct to the IMC, using the APIX2 high speed digital serial link.

General Media Information

Devices that can be connected to the portable media interface panel include:

- USB mass storage devices (for example a memory stick). Devices must use FAT or FAT32 file format.
- iPod® (iPod Classic, iPod touch, iPhone and iPod Nano are supported - full functionality for older devices cannot be guaranteed). iPod Shuffle functionality cannot be guaranteed.
- Auxiliary device (personal audio, MP3 players).
- Devices with Bluetooth® connectivity. Devices must support A2DP and AVRCP Bluetooth® protocols).

When connecting a portable storage device, select AUX on the TS to select that device input. Depending on the device connected, many of the audio controls on the TS and steering wheel switches can be used.



NOTE:

It is recommended to disconnect an iPod when leaving the vehicle. Failure to do so may result in the iPod battery discharging.

To maximise playback quality, it is recommended to use lossless compression for any media files on iPod. Failing this, it is recommended that compressed files utilise a minimum bitrate of 192 kilobits per second (kbps) (a higher bitrate is strongly recommended for increased quality playback).

For media files delivered via USB or MP3 players, the highest compression rate supported is 320 kbps. If anything less than 128 kbps is used, Digital Signal Processing (DSP) functionality may be lost.

Some MP3 players may have a file system that is not supported by the IMC. To use an MP3 player, it must be set to USB Removable Device or Mass Storage Device mode. Only music that has been added to the device in this mode can be played.

UNIVERSAL SERIAL BUS (USB) 3.0 SIGNAL BOOSTER - INCONTROL TOUCH PRO SYSTEMS ONLY

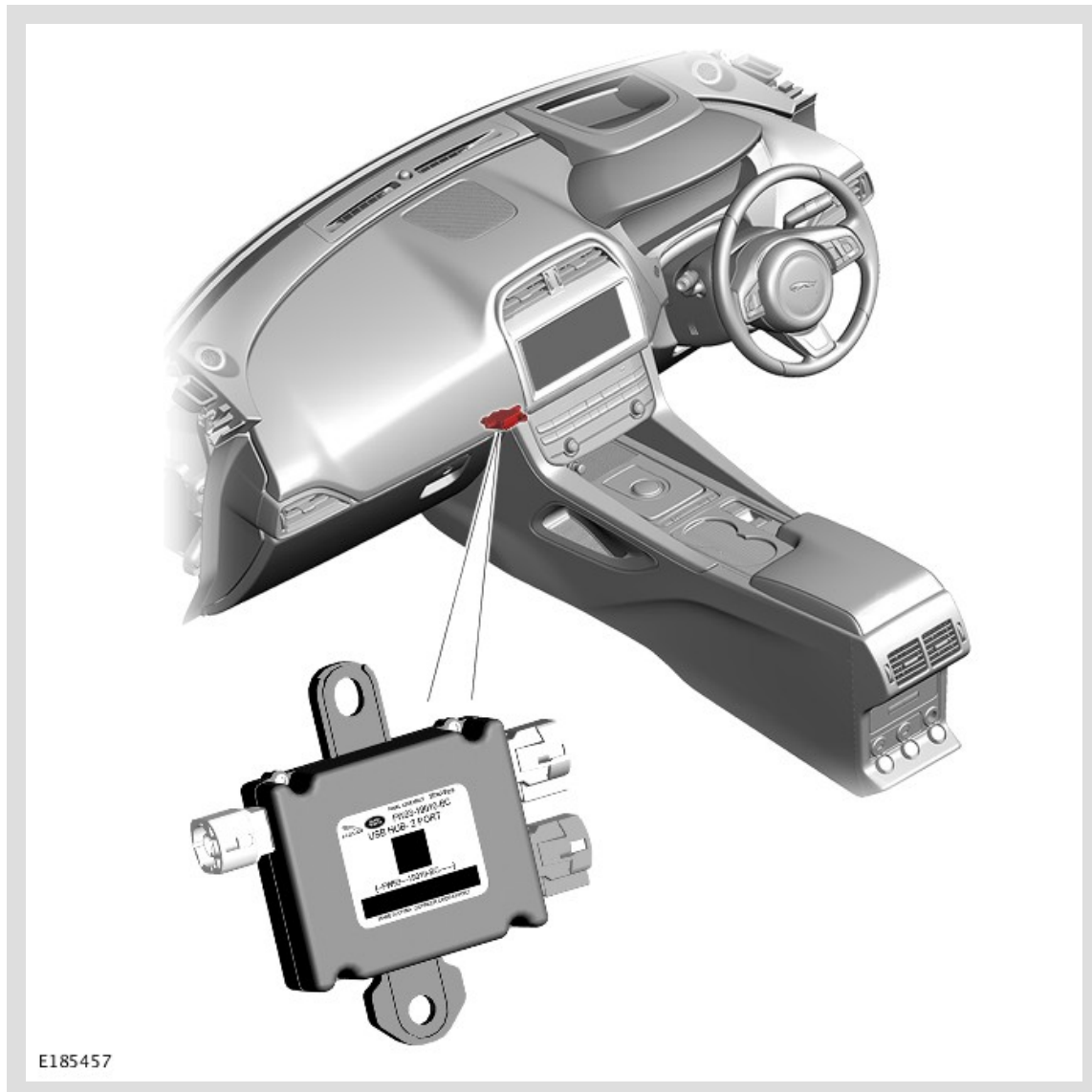


The USB 3.0 signal booster is located on the left side of the rear seat cushion, behind the seat bolster trim.

USB 3.0 signal booster provides data transmission speeds which are 10 times faster than USB 2.0 (480 Mbps). USB 3.0 signal booster significantly reduces the time required for data transmission, reduces power consumption, and is backwards compatible with USB 2.0.

To provide greater data transmission distance, without signal degradation, a USB 3.0 signal booster is fitted in the data transmission line. The signal booster essentially 'amplifies' the signals transmitted between the portable media interface panel USB sockets and the IMC.

UNIVERSAL SERIAL BUS (USB) 2.0 HUB



The USB 2.0 hub is located in the instrument panel, behind the Integrated Control Panel (ICP).

The USB 2.0 hub operates as a signal splitter between the IMC, digital disc player and the electronic toll collection module (Japan only). The USB 2.0 hub also functions as a signal booster for USB 2.0 signals between the connected components.

ANTENNAS

AM, FM, digital radio, satellite radio, Television (TV) and Fuel Fired Booster Heater (FFBH) antennas are located in the roof pod and rear quarter windows. For additional information, refer to: Antenna (415-01 Information and Entertainment System, Description and Operation).

SPEAKERS

The number, location and specification of the speakers in the vehicle differs depending on the level of audio system fitted.

For details of the speaker systems, refer to the 'Speakers' section. For additional information, refer to: Speakers (415-01 Information and Entertainment System, Description and Operation).

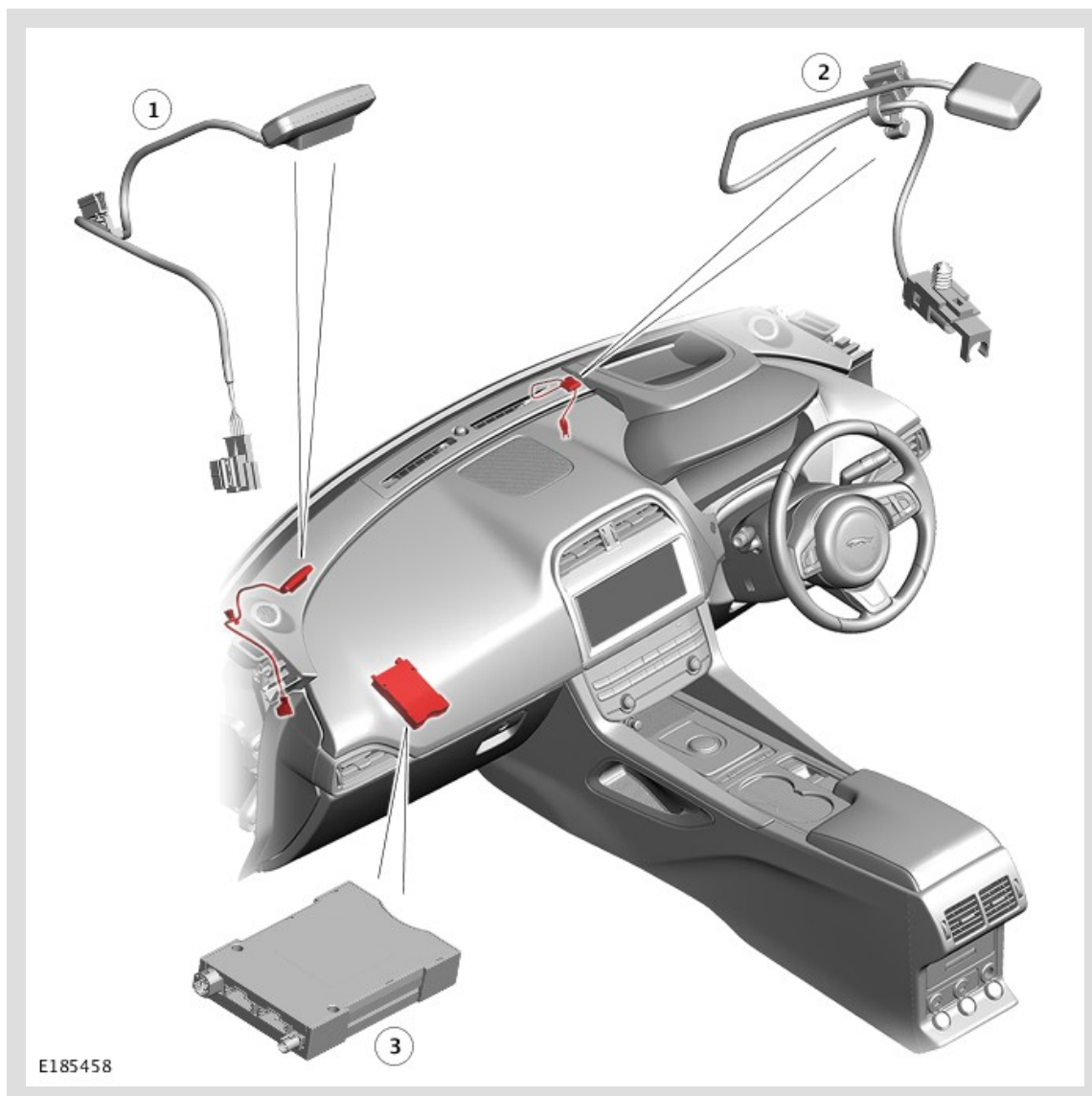
ADVANCED VOICE CONTROL

The advanced voice system provides the driver with the option of voice control for a range of supported functions. The following systems include Voice functionality:

- Radio, satellite radio and digital radio (if fitted)
- CD player
- Hard disk drive/virtual CD player
- USB and auxiliary connections
- Navigation system
- Bluetooth® Telephone system
- Vehicle notepad.

For details of voice control, refer to the 'Voice Control' section. For additional information, refer to: Voice Control (415-01 Information and Entertainment System, Description and Operation).

ELECTRONIC TOLL COLLECTION MODULE (JAPAN ONLY)



ITEM	DESCRIPTION
1	Vehicle Information and Communication System (VICS) beacon antenna
2	Electronic toll collection module antenna
3	Electronic toll collection module

In the Japanese markets, vehicles with the InControl Touch Pro audio system can have an Electronic Toll Collection Module fitted into the audio system.

The electronic toll collection system comprises an electronic toll collection module and a Dedicated Short Range Communication (DSRC) antenna. The antenna is located on the top of the instrument panel on the right side. The electronic toll collection module is located in the glovebox.

An electronic toll collection system credit card is inserted into the electronic toll collection module which reads the data on the card. The electronic toll collection module is connected to the electronic toll collection module antenna which wirelessly communicates via a Dedicated Short Range Communication (DSRC) system with the antenna located at the toll gates. The electronic toll collection module sends and receives vehicle information necessary for paying the appropriate toll fees.

Dedicated Short Range Communication (DSRC) is a system which makes bi-directional communication, within a short distance via radio beacons, on the 5.8 GHz wireless band. DSRC communications enable high-speed, high-capacity data transmission and reception. The system also features high-level security to ensure the safety of the card user's private information.

The electronic toll collection module receives a fused power supply via a relay in the Passenger Junction Box (PJB). The electronic toll collection module antenna is connected directly to the module. The VICS beacon antenna is also connected directly to the module on vehicles fitted with the electronic toll collection system.

The electronic toll collection module is connected to the USB 2.0 hub and can pass relevant information and VICS beacon antenna signals directly to the IMC.

OPERATION

INCONTROL TOUCH SYSTEMS

A permanent primary battery supply is provided to the Audio Head Unit (AHU) from the Battery Junction Box (BJB), via the Quiescent Current Control Module (QCCM).

The QCCM receives battery status signals transmitted by the Battery Monitoring System (BMS) control module. The QCCM can disable the power supply to other non-essential control modules to avoid excessive discharge of the battery. The systems supplied via the QCCM are the infotainment systems and the climate control systems. For additional information, refer to: Battery and Cables (414-01 Battery, Mounting and Cables, Description and Operation).

The AHU is connected to the touch screen via an Automotive Pixel Link 2 (APIX2) connection. APIX2 is a high speed data transmission over a single twisted pair cable. Touch screen selections by the driver are passed to the AHU on the APIX2 connection for processing.

The portable audio interface panel is connected to the AHU with USB data links. The USB sockets provide a 5V charging output for the connected device and also provide external connection of a device to the AHU.

INCONTROL TOUCH PRO SYSTEMS

A permanent primary battery supply is provided to the Infotainment Master Controller (IMC) from the Battery Junction Box (BJB), via the Quiescent Current Control Module (QCCM).

The QCCM receives battery status signals transmitted by the Battery Monitoring System (BMS) control module. The QCCM can disable the power supply to other non-essential control modules to avoid excessive discharge of the battery. The systems supplied via the QCCM are the infotainment systems and the climate control systems. For additional information, refer to: Battery and Cables (414-01 Battery, Mounting and Cables, Description and Operation).

The IMC is connected to the touch screen via an Automotive Pixel Link 2 (APIX2) connection. APIX2 is a high speed data transmission over a single twisted pair cable. Touch screen selections by the driver are passed to the AHU on the APIX2 connection for processing.

The portable media interface panel is connected to the IMC with USB data links and an APIX2 connection. The USB sockets provide a 5V charging output for the connected device and also provide external connection of a device to the IMC. The APIX2 connection provides video input into the IMC from an external device.

Where fitted, the digital disc player is connected to the IMC using a USB connection which is routed via the USB 2.0 hub. Power supply and ground for the digital disc player is supplied from the IMC to the digital disc player via the USB 2.0 hub.

Where fitted, the Audio Amplifier Module (AAM) provides power supply and ground for the headphone transmitter. Data is supplied to the headphone transmitter via twisted pair cable.

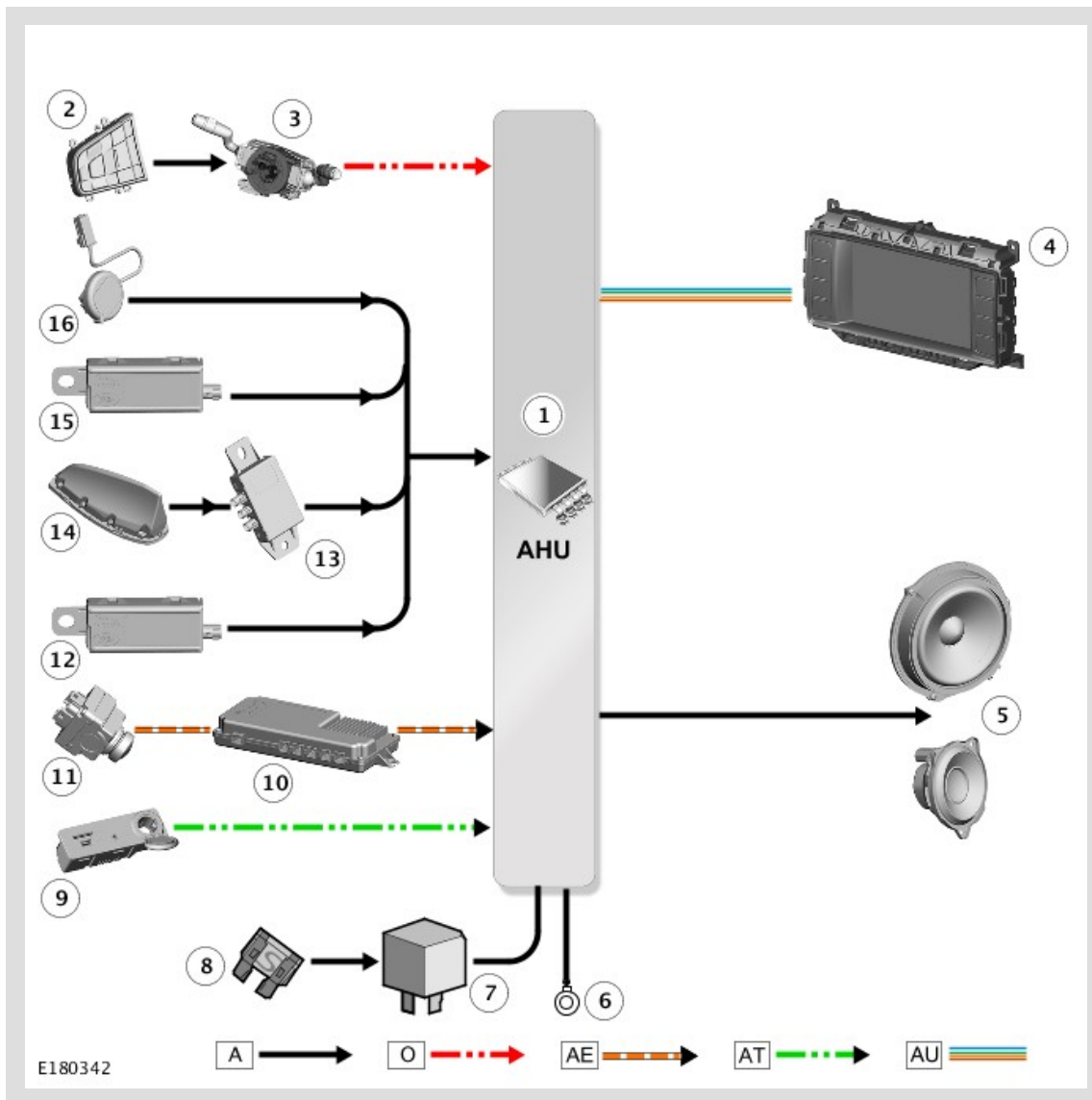
Where the virtual display Instrument Cluster (IC) is fitted, video data is passed from the IMC to the IC on an APIX2 data connection. The data connection provides video output from the IMC for user interface information to be displayed in the IC.

CONTROL DIAGRAM

CONTROL DIAGRAM - INCONTROL TOUCH 8 SPEAKER SYSTEM

 NOTE:

A = Hardwired; **O** = Local Interconnect Network (LIN); **AT** = Universal Serial Bus (USB); **AU** = APIX2



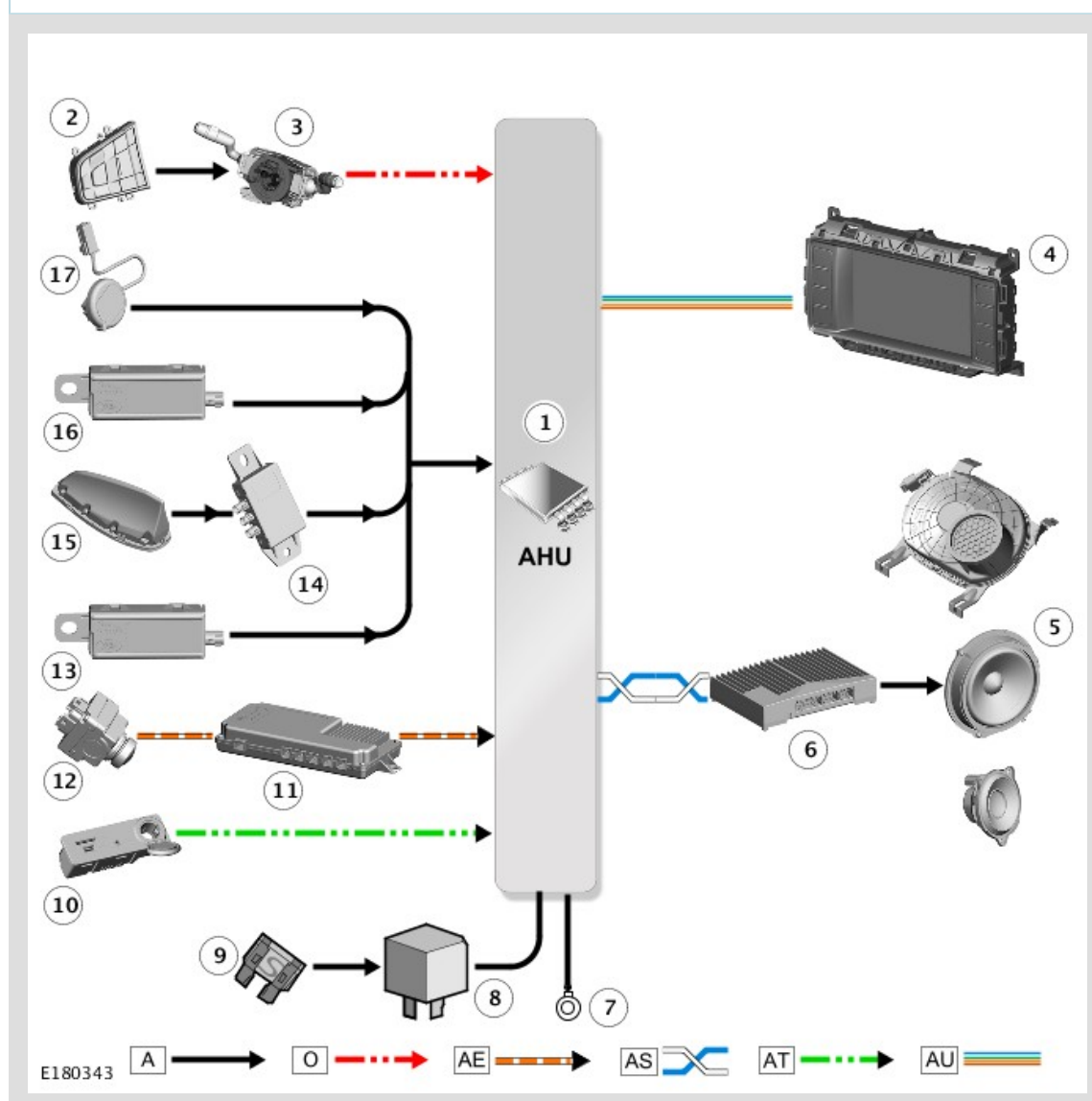
ITEM	DESCRIPTION
1	Audio Head Unit (AHU)
2	Left steering wheel switchpack
3	Clockspring
4	Touch Screen (TS)
5	Vehicle speakers
6	Ground
7	Relay - located in Quiescent Current Control Module (QCCM)

8	Fuse - located in QCCM
9	Portable audio interface panel
10	Camera Control Module (CCM)
11	Rear view camera
12	FM / TMC / DAB-III antenna amplifier
13	GPS signal splitter - if fitted
14	Roof pod - Global Positioning System (GPS) antenna / DAB-L antenna
15	Amplitude Modulation (AM)/FM antenna amplifier
16	Microphone
16	Microphone

CONTROL DIAGRAM - INCONTROL TOUCH MERIDIAN® SOUND 11 SPEAKER SYSTEM

△ NOTE:

A = Hardwired; **O** = Local Interconnect Network (LIN); **AS** = High speed CAN infotainment systems bus; **AT** = Universal Serial Bus (USB); **AU** = APIX2

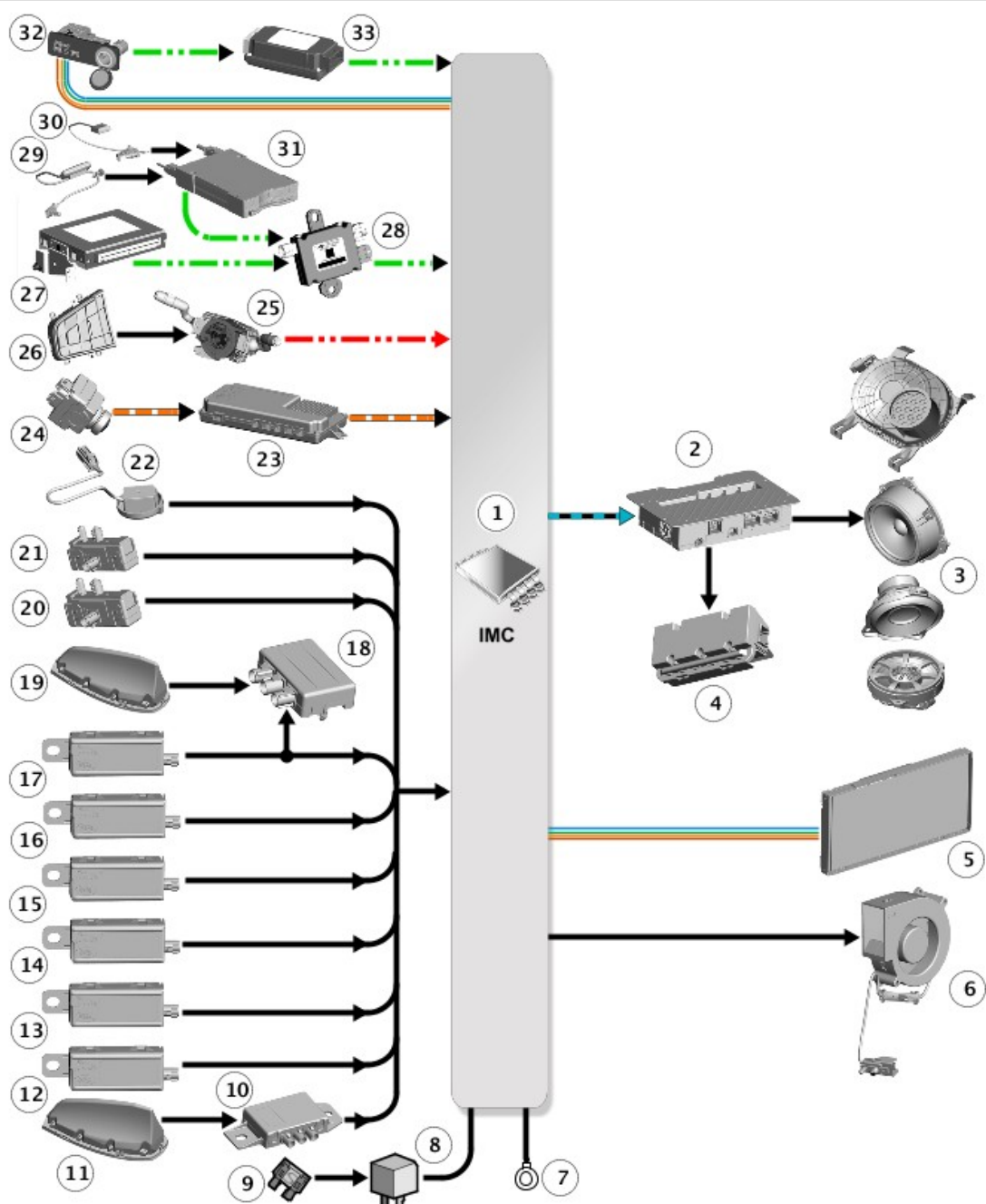


1	Audio Head Unit (AHU)
2	Left steering wheel switchpack
3	Clockspring
4	Touch Screen (TS)
5	Vehicle speakers
6	Audio Amplifier Module (AAM)
7	Ground
8	Relay - located in Quiescent Current Control Module (QCCM)
9	Fuse - located in QCCM
10	Portable audio interface panel
11	Camera Control Module (CCM)
12	Rear view camera
13	FM / TMC / DAB-III antenna amplifier
14	GPS signal splitter - if fitted
15	Roof pod - Global Positioning System (GPS) antenna / DAB-L antenna
16	Amplitude Modulation (AM)/FM antenna amplifier
17	Microphone

CONTROL DIAGRAM - INCONTROL TOUCH PRO MERIDIAN® DIGITAL SOUND 11 SPEAKERS AND MERIDIAN® DIGITAL SURROUND SOUND 17 SPEAKERS SYSTEMS

NOTE:

A = Hardwired; **O** = Local Interconnect Network (LIN); **AE** = LVDS; **AT** = Universal Serial Bus (USB); **AU** = APIX2; **AW** = BroadR-Reach® ethernet



ITEM	DESCRIPTION
1	Infotainment Master Controller (IMC)
2	Audio Amplifier Module (AAM)
3	Vehicle speakers
4	Headphone transmitter
5	Touch Screen (TS)
6	IMC cooling fan
7	Ground
8	Relay - located in Quiescent Current Control Module (QCCM)
9	Fuse - located in QCCM
10	GPS signal splitter
11	Roof pod - GNSS antenna / DAB-L antenna / GSM antenna / SDARS antenna (NAS only)
12	DAB-3 antenna amplifier
13	AM / FM antenna amplifier
14	TV antenna
15	TV antenna
16	TV antenna
17	FM2 / TMC / DAB-3 antenna amplifier
18	Roof pod DAB-L antenna
19	DAB signal combiner
20	Right WiFi antenna
21	Left WiFi Bluetooth™ antenna
22	Microphone
23	Camera Control Module (CCM)
24	Rear view camera
25	Clockspring
26	Left steering wheel switchpack
27	Digital disc player
28	USB 2.0 hub
29	Vehicle Information and Communication System (VICS) beacon antenna (Japan only)
30	Electronic Toll Collection Module antenna
31	Electronic Toll Collection Module
32	Portable media interface panel
33	USB 3.0 signal booster

